

Bachelor's Thesis

Cheap Bootstrap

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Abstract

This thesis examines the bootstrap method, with particular emphasis on the Cheap Bootstrap as an approach for estimating statistical uncertainty, especially confidence intervals. By attaining valid coverage probabilities using only a single bootstrap replication, the Cheap Bootstrap presents a computationally efficient alternative to conventional bootstrap methods, leading to substantial reductions in computational costs. The theoretical foundations of the bootstrap are reviewed, including standard error estimation, commonly used bootstrap confidence interval methods, such as the normal and percentile approaches, and consistency properties. Building on this foundation, the Cheap Bootstrap is introduced, including its confidence interval and the underlying theoretical principles. A comprehensive simulation study evaluates the performance of the Cheap Bootstrap across various scenarios, including normally and gamma-distributed data, extreme quantiles, small sample sizes, and classical linear regression models, comparing its empirical coverage and interval width with those of the normal and percentile bootstrap intervals. The results demonstrate that the Cheap Bootstrap achieves high empirical coverage with minimal bootstrap replications across nearly all tested settings, except for extreme quantiles, and consistently outperforms classical bootstrap methods.

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1 Introduction

A primary goal of inferential statistics is to draw conclusions about an underlying population based on the limited amount of data obtained from a sample. The population of interest is typically a large group of individuals, such as all residents of a given region or all high school students in a country. For these individuals, information such as height or Abitur grade is collected in order to estimate the corresponding population parameters, for example, the mean height and the mean Abitur grade. However, obtaining that information for every individual in a population is often expensive and time-consuming. In practice, we therefore randomly select a subset of individuals from the population without replacement so that each individual is drawn only once. This subset represents a sample from the population and can be used to estimate the parameter of interest. Formally, the unknown parameter of interest, denoted by θ , is estimated using the statistic $\hat{\theta}_n$ calculated from the sample. This setup naturally raises the question of how we can assess the accuracy and precision of the statistic $\hat{\theta}_n$, since it generally differs from the true parameter unless the sample includes every individual. To illustrate this, consider the sample mean. If we could repeatedly draw all possible independent samples of the same size from the population and compute the mean for each of these samples, the resulting values would differ. The distribution of all possible values of the sample mean is referred to as the sampling distribution of the statistic. This distribution allows us to capture the variability of the estimator and to assess both the accuracy and precision of the statistic by computing quantities such as the standard error and confidence intervals. However, in practice, we cannot repeatedly draw samples from the population, and the true sampling distribution of the statistic is therefore unknown. To overcome this limitation, Efron (1979) introduced the **bootstrap**, a data-based simulation method that uses only the observed sample to approximate the sampling distribution of a statistic $\hat{\theta}_n$ under the unknown population distribution F via the bootstrap distribution $H_{\text{boot}}(x)$. This is done by generating “new” samples from the observed dataset with replacement and calculating the statistic for each bootstrap sample to form a set of bootstrap replications. That set serves as an approximation of the sampling distribution, allowing us to compute quantities such as the standard error, bias, and confidence intervals. Typical examples of statistics frequently discussed in the bootstrap literature include the mean, median, or regression coefficients (Efron and Gong, 1983, Efron and Tibshirani, 1986, 1994). For more practical applications of the bootstrap, Davison and Hinkley (1997) provides numerous examples, while Chernick (2011) offers examples that include implementations in R. Theoretical results on the bootstrap, including consistency and asymptotic distributions, are presented in Shao and Tu (1995), Vaart (1998), Bickel and Freedman (1981), and Singh (1981).

While the bootstrap provides a powerful and flexible method for assessing statistical uncertainty, it can become computationally demanding, particularly in modern high-dimensional settings and large-scale models, where the number of features and parameters may be comparable to or even exceed the sample size. Constructing bootstrap confidence intervals typically requires a large number of bootstrap replications, often on the order of hundreds or thousands. Each replication involves resampling the data and refitting potentially complex, large-scale models on every bootstrap sample. As a result, the

computational cost of classical bootstrap methods can become enormous. To address this challenge, Lam (2022) introduced the **Cheap Bootstrap**, a method designed to substantially reduce the computational burden of bootstrap-based inference. Instead of relying on large numbers of bootstrap replications, the Cheap Bootstrap can achieve valid statistical inference with as few as one bootstrap replication. In many works such as Singh (1981), Bickel and Freedman (1981), and Vaart (1998), it is established that both the original estimator and the bootstrap estimators are asymptotically normally distributed. Lam builds upon this result and further shows that the distributions of the original estimator and any number of bootstrap replications are also asymptotically independent. This key insight forms the theoretical foundation of the Cheap Bootstrap, allowing valid inference even with only one resample.

In the following chapters, we first review the classical bootstrap introduced by Efron, presenting the algorithm for computing standard errors and confidence intervals. We then cover bootstrap consistency and conclude the chapter with a step-by-step description of the algorithm. Building on this foundation, we present the Cheap Bootstrap, including the construction of the confidence interval and the theoretical principles of the method. Finally, the empirical performance of the Cheap Bootstrap is evaluated and compared with classical bootstrap methods, focusing on the empirical coverage and average interval width of the confidence intervals. This evaluation is carried out through simulation studies across different datasets, statistics, and challenging scenarios such as small sample sizes or extreme quantiles. The chapter concludes with an application of the methods to linear regression models, where confidence intervals for the model parameters are computed and assessed in terms of empirical coverage and average interval width.

2 Bootstrap

We aim to assess the accuracy and precision of an estimator $\hat{\theta}_n$ for the true parameter θ based on an observed sample. The first approach is to estimate the standard error σ_F of $\hat{\theta}_n$ using the bootstrap method. The second approach is to construct a confidence interval for θ using two different bootstrap methods. Finally, we generalize the approach by approximating the sampling distribution of the statistic with the bootstrap distribution and explaining the consistency of this approximation.

2.1 Bootstrap Algorithm

2.1.1 Bootstrap Standard Error Estimation

Suppose we observe an independent and identically distributed (i.i.d.) random sample

$$\mathbf{X} = (X_1, \dots, X_n) \stackrel{\text{i.i.d.}}{\sim} F, \quad (1)$$

where F is an unknown population distribution:

$$F(x) = P(X \leq x). \quad (2)$$

Let $\mathbf{x} = (x_1, \dots, x_n)$ denote the observed values of the sample. Our aim is to estimate the true unknown parameter denoted by $\theta = \theta(F)$, which is a statistical functional of F . Since F is unknown, θ cannot be computed directly and is estimated by a statistic $\hat{\theta}_n = g(X_1, \dots, X_n)$ for some function g . For example, if $\theta(F)$ denotes the population mean, then the corresponding statistic $\hat{\theta}_n$ is the sample mean computed from the observed data. After computing $\hat{\theta}_n$, we wish to assess its accuracy by estimating the standard error (Efron and Tibshirani, 1986), which is defined as

$$\sigma_F(\hat{\theta}_n) = \sqrt{\text{Var}_F(\hat{\theta}_n)}. \quad (3)$$

This is the standard deviation of the statistic $\hat{\theta}_n$ under F . It is important to note that estimating the standard error is challenging, since for many statistics, except for the sample mean, there simply is no closed-form formula for $\sigma_F(\hat{\theta}_n)$ (Efron and Tibshirani, 1994). To overcome this problem, we approximate $\sigma_F(\hat{\theta}_n)$ using the following bootstrap algorithm, which can be applied to virtually any statistic.

First, we define $\hat{F}_n$ as the empirical distribution function (EDF), which serves as an estimate of the unknown distribution F . The EDF puts probability $1/n$ on each observed value x_i , $i = 1, \dots, n$, and is given by

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x), \quad (4)$$

where the indicator function $I(X_i \leq x)$ is defined as

$$I(X_i \leq x) = \begin{cases} 1, & \text{if } X_i \leq x, \\ 0, & \text{if } X_i > x. \end{cases} \quad (5)$$

Next, we generate a bootstrap sample $\mathbf{X}^* = (X_1^*, \dots, X_n^*)$, which is an i.i.d. random sample of size n drawn from $\hat{F}_n$:

$$X_1^*, \dots, X_n^* \stackrel{\text{i.i.d.}}{\sim} \hat{F}_n. \quad (6)$$

For example, suppose we have 10 observations $\{x_1, x_2, \dots, x_{10}\}$ from the original sample. A possible bootstrap sample of size 10 could consist of

$$\{x_1^*, x_2^*, x_3^*, x_4^*, x_5^*, x_6^*, x_7^*, x_8^*, x_9^*, x_{10}^*\} = \{x_2, x_8, x_3, x_2, x_9, x_5, x_{10}, x_1, x_5, x_6\}.$$

Thus, each x_i^* is simply a data point drawn from the original sample with replacement, so some values may appear zero, once, or multiple times. Finally, we compute the bootstrap replication $\hat{\theta}_n^* = g(X_1^*, \dots, X_n^*)$, obtained by applying the same function g to the bootstrap sample $\mathbf{X}^*$. To estimate the standard error in (3), we first apply the plug-in principle. This means that we replace the unknown distribution F with the empirical distribution $\hat{F}_n$, treating it as if it were the true distribution. This results in

$$\sigma_{\hat{F}_n}(\hat{\theta}_n) = \sqrt{\text{Var}_{\hat{F}_n}(\hat{\theta}_n)}, \quad (7)$$

which is the standard deviation of the statistic $\hat{\theta}_n$ under the empirical distribution $\hat{F}_n$. However, even after substituting F with $\hat{F}_n$, computing (7) may be difficult. The bootstrap method uses Monte Carlo simulation to approximate $\sigma_{\hat{F}_n}(\hat{\theta}_n)$. In practice, this means we generate B bootstrap samples from the original data (typically B ranges from 50 to 1000), compute the bootstrap replications $\hat{\theta}_n^{*(b)} = \hat{\theta}_n(X_1^{*(b)}, \dots, X_n^{*(b)})$ for each bootstrap sample $b = 1, \dots, B$ and estimate the standard error (7) using the sample standard deviation of the bootstrap replications, denoted by $\hat{\sigma}_B$:

$$\hat{\sigma}_B = \left(\frac{\sum_{b=1}^B (\hat{\theta}_n^{*(b)} - \bar{\theta}_n^*)^2}{B - 1} \right)^{1/2}, \quad \text{where } \bar{\theta}_n^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_n^{*(b)}. \quad (8)$$

By the law of large numbers (see A.4), the sample standard deviation of the bootstrap replications satisfies

$$\hat{\sigma}_B \xrightarrow{a.s.} \sigma_{\hat{F}_n}(\hat{\theta}_n) \quad \text{as } B \rightarrow \infty. \quad (9)$$

$$\sigma_{\hat{F}_n}(\hat{\theta}_n) \approx \sigma_F(\hat{\theta}_n) \quad (10)$$

In other words, as the number of bootstrap replications grows, the bootstrap estimate $\hat{\sigma}_B$ converges almost surely to the standard deviation of $\hat{\theta}_n$ under the empirical distribution $\hat{F}_n$, which in turn acts as the approximation of the true standard deviation $\sigma_F(\hat{\theta}_n)$ as the sample size n grows (Wasserman, 2006). However, to assess not only the accuracy but also the precision of the estimator, confidence intervals are typically constructed.

2.1.2 Bootstrap Confidence Intervals

After computing $\hat{\sigma}_B$, we can use it together with the statistic $\hat{\theta}_n$ to construct an approximate confidence interval $\mathcal{I}$ for θ :

$$\theta \in \mathcal{I} = \left[\hat{\theta}_n \pm z_{1-\alpha/2} \hat{\sigma}_B \right], \quad (11)$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of a standard normal distribution $N(0, 1)$. This is also called the normal (or standard) interval for θ . For example, a 95% confidence interval for θ is given by $\left[\hat{\theta}_n \pm 1.96 \cdot \hat{\sigma}_B \right]$.

Another approach called the percentile method uses the $\alpha/2$ and $(1 - \alpha/2)$ -quantiles of the bootstrap replications $\hat{\theta}_n^{*(1)}, \dots, \hat{\theta}_n^{*(B)}$:

$$\left[\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^* \right]. \quad (12)$$

More advanced confidence interval methods, such as the bias-corrected and accelerated (BC_a) or the approximate bootstrap confidence (ABC) intervals, have been proposed by DiCiccio and Efron (1996), but these are beyond the scope of this thesis.

2.2 Bootstrap Consistency

Another important aspect besides the accuracy of an estimator is its consistency. The statistic $\hat{\theta}_n$ is called consistent if

$$\hat{\theta}_n \xrightarrow{P} \theta. \quad (13)$$

That is, as the sample size n increases, the statistic $\hat{\theta}_n$ converges in probability to the true parameter θ . While the consistency of $\hat{\theta}_n$ is important, in the bootstrap literature, consistency is often defined and proven for the distribution of the deviation $\hat{\theta}_n - \theta$. Let us first define $H_F(x)$ as the distribution of $\hat{\theta}_n - \theta$ under the unknown distribution F :

$$H_F(x) = P_F \left(\hat{\theta}_n - \theta \leq x \right). \quad (14)$$

Because $H_F(x)$ depends on the unknown distribution F , we need to estimate it using the bootstrap. Therefore, we replace F with the empirical distribution $\hat{F}_n$ and calculate the bootstrap replication $\hat{\theta}_n^*$ from the bootstrap sample drawn from $\hat{F}_n$. The resulting theoretical bootstrap estimator of $H_F(x)$ is

$$H_{boot}(x) = H_{\hat{F}_n}(x) = P_* \left(\hat{\theta}_n^* - \hat{\theta}_n \leq x \mid \hat{F}_n \right) \quad (15)$$

where $P_*(\cdot \mid \hat{F}_n)$ denotes the conditional probability given the empirical distribution $\hat{F}_n$ constructed from the observed data $X_1, \dots, X_n$. We call the bootstrap distribution $H_{boot}(x)$ consistent if

$$\sup_x |H_{boot}(x) - H_F(x)| \xrightarrow{P} 0, \quad \text{as } n \rightarrow \infty. \quad (16)$$

This means that the bootstrap distribution $H_{boot}(x)$ converges in probability to the true distribution $H_F(x)$, with respect to the supremum norm, as the sample size n grows to infinity.

To prove the consistency of $H_{boot}(x)$, we refer to Vaart (1998) and Lam (2022). We assume that the distributions of $(\hat{\theta}_n - \theta)$ and of the bootstrap estimator $(\hat{\theta}_n^* - \hat{\theta}_n)$ both converge to the same continuous distribution G , where the former converges in distribution and the latter converges conditionally in probability. In our setting, G is the normal distribution $N(0, \sigma^2)$. Under this assumption, the distributions in (14) and (15) satisfy

$$P_F(\hat{\theta}_n - \theta \leq x) \xrightarrow{d} G(x) \quad (17)$$

and

$$P_*(\hat{\theta}_n^* - \hat{\theta}_n \leq x \mid \hat{F}_n) \xrightarrow{P} G(x). \quad (18)$$

Here, it is important to distinguish between the two notions of convergence: convergence in distribution and conditional convergence in probability. In (17), we have convergence in distribution, meaning that as the sample size n increases, the distribution of $\hat{\theta}_n - \theta$ increasingly resembles the normal distribution G . In (18), the randomness of the bootstrap distribution comes from the fact that it depends on the observed sample. If we draw a different random sample from the population, the bootstrap distribution would change accordingly. Therefore, conditional on the observed sample, the bootstrap distribution converges in probability to the normal distribution as the sample size increases. To illustrate why they both converge to the same normal distribution, consider the sample mean as our given statistic. We denote the sample mean by

$$\hat{\theta}_n = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad (19)$$

where $X_1, \dots, X_n$ are i.i.d. drawn from the unknown distribution F with a finite mean μ and variance σ^2 . Based on the Central Limit Theorem, the deviation of the sample mean from the true mean, scaled by $\sqrt{n}$, converges in distribution to a normal distribution. More precisely,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2). \quad (20)$$

Now consider the bootstrap sample $X_1^*, \dots, X_n^*$ drawn with replacement from the original data set, and let

$$\hat{\theta}_n^* = \bar{X}_n^* = \frac{1}{n} \sum_{i=1}^n X_i^* \quad (21)$$

be the corresponding bootstrap sample mean. We now establish the bootstrap analogue of the result in (20) for the sequence $\sqrt{n}(\bar{X}_n^* - \bar{X}_n)$. Applying a univariate version of Theorem 23.4 in Vaart (1998), we conclude that the bootstrap sequence converges asymptotically to a normal distribution:

Theorem 2.1 (Sample Mean). *Let $X_1, X_2, \dots$ be i.i.d. random variables with mean μ and variance σ^2 . Then conditionally on $X_1, X_2, \dots$, for almost every sequence $X_1, X_2, \dots$,*

$$\sqrt{n} (\bar{X}_n^* - \bar{X}_n) \xrightarrow[*]{d} N(0, \sigma^2).$$

The proof of this theorem can be found in Vaart (1998), Chapter 23. We now extend it to a general statistic $\hat{\theta}_n$. Following Lam (2022), we make the standard assumption for bootstrap validity:

Assumption 2.1 (Standard condition for bootstrap validity). *We have $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ where $\sigma^2 > 0$. Moreover, a bootstrap replication $\hat{\theta}_n^*$ satisfies $\sqrt{n}(\hat{\theta}_n^* - \hat{\theta}_n) \xrightarrow[*]{d} N(0, \sigma^2)$ conditional on the data $X_1, X_2, \dots$ in probability as $n \rightarrow \infty$.*

Assumption 2.1 is satisfied when the parameter θ is Hadamard differentiable (the definition can be found in Shao and Tu (1995), Chapter 2) with respect to F . Under this assumption, the consistency condition (16) follows: the bootstrap distribution $H_{boot}(x)$ converges in probability, conditional on the data, to the true sampling distribution $H_F(x)$. Consequently, the bootstrap distribution, which can be computed using the algorithm described in Chapter 2.1, yields a valid approximation of the sampling distribution of the statistic.

2.3 Step-by-Step Bootstrap Algorithm

To summarize the steps of the bootstrap procedure, we provide the algorithm in pseudocode below.

Algorithm 1 Bootstrap Resampling Algorithm

Input: Original sample $\mathbf{X} = (X_1, \dots, X_n) \stackrel{\text{i.i.d.}}{\sim} F$, number of bootstrap replications B , statistical functional $\hat{\theta}_n = g(\cdot)$

Output: Bootstrap estimate $\hat{\sigma}_B$ for standard error of $\hat{\theta}_n$ and confidence intervals for θ

- 1: Compute the original statistic: $\hat{\theta}_n = g(\mathbf{X})$
- 2: Construct empirical distribution function (EDF):

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x)$$

- 3: **for** $b = 1, \dots, B$ **do**
- 4: Draw a bootstrap sample $\mathbf{X}^{*b} = (X_1^{*b}, \dots, X_n^{*b})$ of size n from $\hat{F}_n$ (sampling with replacement from $\mathbf{X}$)
- 5: Compute bootstrap replication $\hat{\theta}_n^{*b} = g(\mathbf{X}^{*b})$
- 6: **end for**
- 7: Compute bootstrap estimate $\hat{\sigma}_B$ for $\sigma_{\hat{F}_n}(\hat{\theta}_n)$:

$$\hat{\sigma}_B = \sqrt{\frac{1}{B-1} \sum_{b=1}^B \left(\hat{\theta}_n^{*b} - \bar{\theta}_n^* \right)^2}, \quad \bar{\theta}_n^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_n^{*b}$$

- 8: Construct normal confidence interval:

$$\left[\hat{\theta}_n - z_{1-\alpha/2} \hat{\sigma}_B, \hat{\theta}_n + z_{1-\alpha/2} \hat{\sigma}_B \right]$$

3 Cheap Bootstrap

3.1 Cheap Bootstrap Confidence Interval

As discussed in the previous chapter, the bootstrap method allows us to estimate a statistic's standard error, construct confidence intervals for the underlying parameter, and approximate the statistic's sampling distribution. The method involves generating many bootstrap samples (e.g. $B = 50, 100, 250, 1000$) and computing the corresponding bootstrap replications $\hat{\theta}_n^{*b}$ for each sample $b = 1, \dots, B$. For classical statistical problems where the parameter of interest may be the mean or median, the computational cost of the bootstrap is relatively small. However, in modern large-scale settings where each resampling step requires refitting a model, the computational cost can become very high.

To address this issue, Lam (2022) introduced the **Cheap Bootstrap**, which drastically reduces the computational cost of bootstrapping for complex models. Rather than generating hundreds or even thousands of bootstrap replications, Lam's method requires only a single bootstrap replication to obtain valid inferential results. In practice, to apply the Cheap Bootstrap, we follow the same steps in Algorithm 1, but set $B = 1$ and construct a confidence interval directly from a single bootstrap replication. In contrast to the standard interval, which requires at least $B \geq 2$ bootstrap replications to estimate the standard error $\hat{\sigma}_B$ needed for interval construction, the Cheap Bootstrap interval is well defined for any positive integer $B \geq 1$:

$$\mathcal{I} = \left[\hat{\theta}_n - t_{B,1-\alpha/2} S, \hat{\theta}_n + t_{B,1-\alpha/2} S \right] \quad (22)$$

where

$$S^2 = \frac{1}{B} \sum_{b=1}^B \left(\hat{\theta}_n^{*b} - \hat{\theta}_n \right)^2. \quad (23)$$

In the case of $B = 1$, the interval (22) simplifies to:

$$\left[\hat{\theta}_n - t_{1,1-\alpha/2} \left| \hat{\theta}_n^* - \hat{\theta}_n \right|, \hat{\theta}_n + t_{1,1-\alpha/2} \left| \hat{\theta}_n^* - \hat{\theta}_n \right| \right]. \quad (24)$$

As B becomes large, the Cheap Bootstrap interval and the standard interval become increasingly comparable. S is similar to the bootstrap standard error $\hat{\sigma}_B$, but it differs in two aspects: first, the deviations are calculated relative to the original statistic $\hat{\theta}_n$ rather than the mean of the bootstrap replications $\bar{\theta}_n^*$, and second, the denominator is B instead of $B - 1$. The critical value $t_{B,1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of t_B , the Student t -distribution with degree of freedom B . As $B \rightarrow \infty$, $S \approx \hat{\sigma}_{boot}$, and $t_{B,1-\alpha/2} \approx z_{1-\alpha/2}$. Thus, the Cheap Bootstrap extends the standard normal interval to allow interval estimation with as few as one bootstrap replication, while still obtaining valid inferential results. The reasoning behind this will be explained in the following section.

3.2 Theory behind the Cheap Bootstrap Interval

We revisit Assumption 2.1 from Chapter 2.2, which ensures the validity of the bootstrap. Under this assumption, both $\sqrt{n}(\hat{\theta}_n - \theta)$ and $\sqrt{n}(\hat{\theta}_n^* - \hat{\theta}_n)$ converge either in distribution or conditionally in probability to the same normal distribution. Using this assumption, the Cheap Bootstrap interval (22) is asymptotically exact. The following theorem, proposed by Lam (2022), formally establishes this property.

Theorem 3.1 (Asymptotic Exactness of the Cheap Bootstrap). *Under Assumption 2.1, for any $B \geq 1$, the interval $\mathcal{I}$ in (22) is an asymptotically exact $(1 - \alpha)$ -level confidence interval for θ :*

$$\mathbb{P}_n(\theta \in \mathcal{I}) \longrightarrow 1 - \alpha \quad \text{as } n \rightarrow \infty,$$

where $\mathbb{P}_n$ denotes the probability with respect to both the data $X_1, \dots, X_n$ and all randomness from the resampling.

The theorem implies that $\mathcal{I}$ achieves asymptotically exact coverage under the standard conditions for bootstrap validity, even when only a single bootstrap replication is used. The asymptotically exact coverage follows from the fact that the estimator and its bootstrap replications are jointly asymptotically independent and normally distributed. This joint asymptotic independence and normality is formally stated in the following proposition, as demonstrated by Lam (2022).

Proposition 1 (Asymptotic Independence and Normality). *Under Assumption 2.1, let $\hat{\theta}_n$ be the original statistic, and let $\hat{\theta}_n^{*b}$, $b = 1, \dots, B$ be the bootstrap replications. Then*

$$\sqrt{n}(\hat{\theta}_n - \theta, \hat{\theta}_n^{*1} - \hat{\theta}_n, \dots, \hat{\theta}_n^{*B} - \hat{\theta}_n) \xrightarrow{d} (\sigma Z_0, \sigma Z_1, \dots, \sigma Z_B),$$

where $Z_0, \dots, Z_B$ are i.i.d. $N(0, 1)$ random variables.

Proof of Proposition 1. Let Φ_σ denote the distribution function of $N(0, \sigma^2)$. We aim to show that for any $x_0, \dots, x_B \in \mathbb{R}$, the joint distribution of $\sqrt{n}(\hat{\theta}_n - \theta)$ and $\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n)$, $b = 1, \dots, B$ converges weakly to the product of i.i.d $N(0, \sigma^2)$ distributions.

$$P\left(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0, \sqrt{n}(\hat{\theta}_n^{*1} - \hat{\theta}_n) \leq x_1, \dots, \sqrt{n}(\hat{\theta}_n^{*B} - \hat{\theta}_n) \leq x_B\right) \longrightarrow \prod_{b=0}^B \Phi_\sigma(x_b) \quad (25)$$

as $n \rightarrow \infty$.

Step 1: Rewrite the joint distribution as an expectation of a product of indicator functions.

$$\mathbb{E}\left[I\left(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0\right) I\left(\sqrt{n}(\hat{\theta}_n^{*1} - \hat{\theta}_n) \leq x_1\right) \cdots I\left(\sqrt{n}(\hat{\theta}_n^{*B} - \hat{\theta}_n) \leq x_B\right)\right].$$

Step 2: Apply the law of iterated expectations. Conditional on the empirical distribution $\hat{F}_n$, the bootstrap replications $\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n)$ are independent. Hence, we can write:

$$\mathbb{E}\left[\mathbb{E}\left[I\left(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0\right) I\left(\sqrt{n}(\hat{\theta}_n^{*1} - \hat{\theta}_n) \leq x_1\right) \cdots I\left(\sqrt{n}(\hat{\theta}_n^{*B} - \hat{\theta}_n) \leq x_B\right) \mid \hat{F}_n\right]\right].$$

Step 3: Rewrite as a product of conditional expectations. Exploiting the conditional independence of the bootstrap replications, the inner expectation can be factorized as follows:

$$\mathbb{E} \left[\mathbb{E} \left[I(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) \mid \hat{F}_n \right] \prod_{b=1}^B \mathbb{E} \left[I(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b) \mid \hat{F}_n \right] \right].$$

Step 4: The first conditional expectation is fixed given the empirical distribution. Note that the indicator $I(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0)$ does not depend on the empirical distribution $\hat{F}_n$, but only on the original sample. Hence, it is fixed given $\hat{F}_n$ and its conditional expectation coincides with the indicator itself:

$$\mathbb{E} \left[I(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) \mid \hat{F}_n \right] = I(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0).$$

Step 5: Express the remaining conditional expectations as probabilities. The remaining indicators $I(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b)$ for $b = 1, \dots, B$ do depend on the empirical distribution $\hat{F}_n$. Therefore, their conditional expectations can be expressed as conditional probabilities:

$$\mathbb{E} \left[I(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b) \mid \hat{F}_n \right] = P(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b \mid \hat{F}_n).$$

Step 6: Combine the previous steps. Putting the above steps together and subtracting the product of normal distribution functions (25), we arrive at the following expression:

$$\left| \mathbb{E} \left[I(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) \prod_{b=1}^B P(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b \mid \hat{F}_n) \right] - \prod_{b=0}^B \Phi_\sigma(x_b) \right|. \quad (26)$$

This expression will be used to establish the joint asymptotic independence and normality.

Step 7: Split the expression into two parts. We want to split (26) into two parts by adding and subtracting the term $P(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) \prod_{b=1}^B \Phi_\sigma(x_b)$. This yields:

$$\underbrace{\left| \mathbb{E} \left[I(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) \prod_{b=1}^B P(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b \mid \hat{F}_n) \right] - P(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) \prod_{b=1}^B \Phi_\sigma(x_b) \right|}_{\text{First term}} + \underbrace{\left| P(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) \prod_{b=1}^B \Phi_\sigma(x_b) - \prod_{b=0}^B \Phi_\sigma(x_b) \right|}_{\text{Second term}}.$$

Step 8: Factorize both terms and apply the triangle inequality. We now factorize the two parts obtained in Step 7 and simultaneously apply the triangle inequality

to separate the two parts:

$$\begin{aligned} &\leq \mathbb{E} \left[I(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) \left| \prod_{b=1}^B P(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b \mid \hat{F}_n) - \prod_{b=1}^B \Phi_\sigma(x_b) \right| \right] \\ &\quad + \left| \left(P(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) - \Phi_\sigma(x_0) \right) \prod_{b=1}^B \Phi_\sigma(x_b) \right|. \end{aligned} \quad (27)$$

Step 9: Replace the indicator function with 1 to obtain an upper bound.

In the first term of Step 8, we know that the indicator function only takes values 0 or 1, so $I(\cdot) \leq 1$. This gives an upper bound, allowing us to simplify the previous expression to

$$\begin{aligned} &\leq \mathbb{E} \left| P(\sqrt{n}(\hat{\theta}_n^{*1} - \hat{\theta}_n) \leq x_1 \mid \hat{F}_n) \cdots P(\sqrt{n}(\hat{\theta}_n^{*B} - \hat{\theta}_n) \leq x_B \mid \hat{F}_n) - \prod_{b=1}^B \Phi_\sigma(x_b) \right| \\ &\quad + \left| P(\sqrt{n}(\hat{\theta}_n - \theta) \leq x_0) - \Phi_\sigma(x_0) \right| \prod_{b=1}^B \Phi_\sigma(x_b). \end{aligned} \quad (28)$$

Based on Assumption 2.1, we know that each bootstrap replication satisfies

$$P(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b \mid \hat{F}_n) \xrightarrow{P} \Phi_\sigma(x_b), \quad b = 1, \dots, B.$$

It then follows that the product of these B conditional probabilities also converges in probability to the product of the corresponding normal distribution functions, i.e.,

$$\prod_{b=1}^B P(\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n) \leq x_b \mid \hat{F}_n) \xrightarrow{P} \prod_{b=1}^B \Phi_\sigma(x_b).$$

To show that the expectation of the difference converges to zero, we note that the terms inside the expectation are bounded by 1, as both conditional probabilities and values of the normal distribution function lie between 0 and 1. Since the difference inside the absolute value converges in probability to zero and the terms are dominated by a constant, the dominated convergence theorem implies that the expectation converges to zero as $n \rightarrow \infty$. For the second term, the absolute difference also vanishes in the limit. Hence, both terms converge to zero, establishing the joint asymptotic independence and normality. $\square$

Now we can prove the asymptotic exactness of Cheap Bootstrap intervals (Theorem 3.1) using Proposition 1. Accordingly, we introduce an approximate pivotal statistic. A pivotal statistic is a quantity whose distribution does not depend on unknown parameters such as the target parameter θ or the nuisance parameter σ . Following Lam, we define the pivotal statistic as

$$T = \frac{\hat{\theta}_n - \theta}{S}, \quad (29)$$

where S^2 is defined in (23). In the following, we show that as the sample size increases, the distribution of T converges in distribution to a Student t -distribution with B degrees of freedom, using Proposition 1 and the continuous mapping theorem (see Theorem A.3):

$$T = \frac{\hat{\theta}_n - \theta}{S} = \frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\sqrt{\frac{1}{B} \sum_{b=1}^B (\sqrt{n}(\hat{\theta}_n^{*b} - \hat{\theta}_n))^2}} \xrightarrow{d} \frac{\sigma Z_0}{\sqrt{\frac{1}{B} \sum_{b=1}^B (\sigma Z_b)^2}} = \frac{Z_0}{\sqrt{\frac{1}{B} \sum_{b=1}^B (Z_b)^2}} \quad (30)$$

where $Z_0, Z_1, \dots, Z_B$ are i.i.d. $N(0, 1)$ variables. Moreover, (30) can then be written as

$$\frac{Z_0}{\sqrt{\frac{1}{B} \sum_{b=1}^B (Z_b)^2}} \stackrel{d}{=} \frac{N(0, 1)}{\sqrt{\chi_B^2/B}} \stackrel{d}{=} t_B \quad (31)$$

where $N(0, 1)$ and χ_B^2 denote the standard normal and chi-squared distributions with B degrees of freedom, respectively. Here, $\stackrel{d}{=}$ indicates equality in distribution. The equalities hold due to the construction of χ_B^2 and t_B : the denominator consists of the sum of squared independent $N(0, 1)$ random variables, which by definition follows a χ^2 distribution with B degrees of freedom. Furthermore, by definition of the t_B distribution, a t_B random variable is a standard normal variable divided by the square root of an independent χ_B^2 variable divided by B . Thus, our pivotal statistic converges in distribution to t_B as $n \rightarrow \infty$. Consequently, the quantiles of the t_B distribution can be used to construct an asymptotically exact confidence interval for θ :

$$\begin{aligned} & \mathbb{P}_n \left(-t_{B, 1-\alpha/2} \leq \frac{\hat{\theta}_n - \theta}{S} \leq t_{B, 1-\alpha/2} \right) \longrightarrow 1 - \alpha \\ \Leftrightarrow & \mathbb{P}_n \left(\hat{\theta}_n - t_{B, 1-\alpha/2} S \leq \theta \leq \hat{\theta}_n + t_{B, 1-\alpha/2} S \right) \longrightarrow 1 - \alpha. \end{aligned} \quad (32)$$

This yields the Cheap Bootstrap confidence interval stated in (22).

4 Simulation Study

4.1 Simulation Setup

In this simulation study, we assess how well the Cheap Bootstrap performs in constructing asymptotically exact confidence intervals for three parameters: the mean, the median, and a p -quantile. We compare the empirical coverage and mean interval width of these intervals with those of the Percentile Bootstrap and the Normal Bootstrap. We also evaluate the performance of the intervals under different distributional shapes. Therefore, for each data-generating process, we generate a random sample of i.i.d. random variables $X_1, \dots, X_n$ and denote the corresponding observed values by $x_1, \dots, x_n$. The first data-generating process is the normal distribution, a continuous and symmetric distribution characterized by the mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, denoted by $X \sim N(\mu, \sigma^2)$. The second data-generating process is the gamma distribution, a continuous and right-skewed distribution characterized by a shape parameter $\alpha > 0$ and a rate parameter $\lambda > 0$, denoted by $X \sim \Gamma(\alpha, \lambda)$.

After generating the samples of size n , we compute three sample statistics that serve as estimators of the corresponding population parameters. First, the **sample mean** is given by

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i,$$

which is the sum of all values divided by the sample size. For the **sample median**, we sort the values in $\mathbf{x}$ in ascending order:

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}.$$

Then the sample median is defined as

$$\hat{x}_{med} = \begin{cases} x_{(\frac{n+1}{2})}, & \text{if } n \text{ is odd} \\ \frac{1}{2} (x_{(n/2)} + x_{(n/2+1)}), & \text{if } n \text{ is even} \end{cases}$$

If the sample size n is odd, the median corresponds to the middle observation of the ordered sample. If n is even, it is the arithmetic mean of the two central observations. Finally, for the **sample p -quantile**, we also start by sorting the sample values in ascending order. The sample p -quantile is then defined as

$$\begin{aligned} \hat{x}_p &= x_{([np]+1)}, & \text{if } np \text{ is not an integer,} \\ \hat{x}_p &\in [x_{(np)}, x_{(np+1)}], & \text{if } np \text{ is an integer,} \end{aligned}$$

where $[np]$ denotes the floor of np . For example, suppose we estimate the p -quantile with $p = 0.25$ based on a sample of size $n = 27$. Then we obtain $np = 27 \cdot 0.25 = 6.75$, which is not an integer. Thus, the floor is $[np] = [6.75] = 6$, and the sample quantile is $\hat{x}_{0.25} = x_{([np]+1)} = x_{(7)}$. This corresponds to the value at position 7 in the ordered sample, such that at least 25% of the sample is less than or equal to $\hat{x}_{0.25}$ and at least 75% of the sample is greater than or equal to $\hat{x}_{0.25}$.

For each distribution, the true parameter values are known, allowing us to assess whether the confidence intervals contain them. For the normal distribution, the true

values of the mean and the median are both equal to μ , while the true p -quantile is given by $x_p = \mu + \sigma z_p$, where z_p denotes the p -quantile of the standard normal distribution. In the simulation study, we set $\mu = 10$, $\sigma = 2$, and $p = 0.75$. For the gamma distribution, the true mean is given by $\frac{\alpha}{\lambda}$. Since neither the median nor the p -quantile has a closed-form expression, these quantities are computed numerically using the inverse cumulative distribution function. In the simulation, we use the `qgamma` function in R to obtain the corresponding values. We set $\alpha = 5$, $\lambda = 2$, and again choose $p = 0.75$.

4.2 Simulation Procedure and Results

Now that the datasets have been generated and the sample statistics and true parameter values have been computed, we evaluate the performance of the confidence intervals using empirical coverage and mean interval width, while also varying the number of bootstrap replications used to construct the intervals. To do this, we apply the bootstrap algorithm described in Chapter 2.3 with different values of B , namely $B = 1, 2, 5, 10, 50, 100$. For each value of B , we generate $N = 1000$ independent datasets of size 100 and construct a 95% confidence interval for each dataset using each bootstrap method. This allows us to assess empirical coverage, which is defined as the proportion of confidence intervals that contain the true parameter.

4.2.1 Results for normally distributed data

We first examine the sample mean. The Cheap Bootstrap attains coverage near the nominal 95%-level with a single bootstrap replication ($B = 1$), yielding an empirical coverage of 94.6%. In contrast, confidence intervals based on the Normal Bootstrap cannot be constructed at $B = 1$, since estimating the bootstrap standard error requires at least two replications (the denominator is zero when $B = 1$). For the Percentile Bootstrap, using a single bootstrap replication results in a degenerate interval, as the lower and upper bounds are equal. For $B = 2$, coverage values for the Normal and Percentile Bootstrap become available. However, both methods perform worse than the Cheap Bootstrap, as the latter attains a coverage probability of 94.7%. In particular, the Percentile Bootstrap performs the worst at 31.5%, which can be linked to the very limited information supplied by only two bootstrap replications. When the number of bootstrap replications used to construct the intervals is increased to $B = 50$ or $B = 100$, all three methods achieve coverage close to the nominal 95%-level (see Figure 1a). As described in Lam (2022), the reduced computational cost of the Cheap Bootstrap comes at the expense of wider confidence intervals. This trade-off is clearly visible in our results. For $B = 1$, the Cheap Bootstrap produces intervals with an average width of 3.87. Increasing the number of bootstrap replications to $B = 2$ leads to a sharp reduction in the average interval width for the Cheap Bootstrap, which decreases from 3.87 to 1.53. Nevertheless, these intervals remain considerably wider than those obtained from the Normal and Percentile Bootstrap, whose average widths are 0.64 and 0.22, respectively. As B increases further, the average interval width of the Cheap Bootstrap continues to decrease and becomes very similar to the average interval widths of the Normal and Percentile Bootstrap for $B = 100$ (see Figure 2a).

We next consider the sample median. Overall, the Cheap Bootstrap method performs slightly worse for the median than for the mean, but it still yields relatively strong results with empirical coverage close to the nominal 95%-level. For $B = 1$, the Cheap Bootstrap achieves an empirical coverage of 90.9%, which increases slightly as B grows. Similar to the sample mean, confidence intervals for the Normal and Percentile Bootstrap become available for $B = 2$, yielding lower coverage values of 68.7% for the Normal and 31.7% for the Percentile Bootstrap. However, as the number of bootstrap replications increases, the coverage of both methods improves steadily, and for $B = 100$, all three methods achieve empirical coverages close to the nominal 95%-level (see Figure 1b). The behavior of the interval widths for the median closely mirrors that observed for the mean. For $B = 1$, the Cheap Bootstrap produces very wide intervals, with an average width of 5.05. However, the average width sharply decreases to 1.97 for $B = 2$; nevertheless, the Cheap Bootstrap intervals remain substantially wider than those obtained from the Normal and Percentile Bootstrap, whose average widths at $B = 2$ are 0.80 and 0.27, respectively. As B increases further, the interval widths of all three methods gradually converge to the same value, as we can see in Figure 2b.

We finally consider the sample quantile. The Cheap Bootstrap again performs slightly worse compared to the mean and the median, but still achieves high empirical coverage. As before, only the Cheap Bootstrap provides meaningful intervals for $B = 1$, while the Normal and Percentile methods require at least two bootstrap replications. As the number of bootstrap replications increases, the empirical coverage of all methods steadily increases toward the nominal 95%-level, as illustrated in Figure 1c. The average interval widths show the same pattern as for the mean and median: they are very wide for $B = 1$, decrease sharply for $B = 2$, and gradually converge to a similar value for all methods as B becomes large (see Figure 2c).

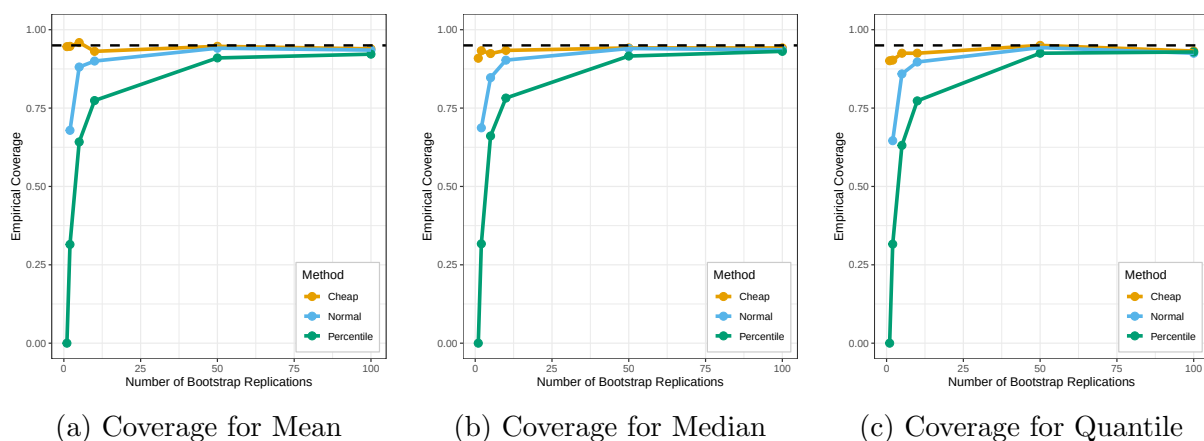


Figure 1: Empirical coverage of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the mean, median, and 0.75-quantile under normally distributed data, based on 1000 simulated intervals for each value of B .

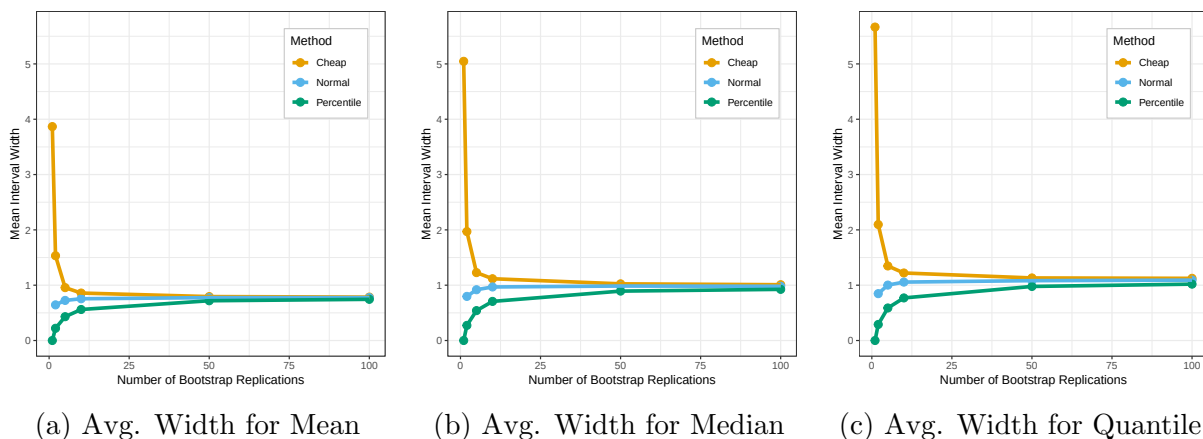


Figure 2: Average widths of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the mean, median, and 0.75-quantile under normally distributed data, based on 1000 simulated intervals for each value of B .

4.2.2 Results for gamma distributed data

We now examine the performance of the Cheap Bootstrap under right-skewed data by considering observations drawn from a Gamma distribution. We again begin with the sample mean. As shown in Figure 3a, the Cheap Bootstrap attains empirical coverage very close to the nominal 95% level for all considered values of B . In comparison, the Normal and Percentile Bootstrap require substantially larger numbers of bootstrap replications ($B = 50$ for Normal and $B = 100$ for Percentile) before achieving performance comparable to that of the Cheap Bootstrap. Numerical results for the empirical coverage of the three methods are reported in Table 3. Similar to our previous simulation study, the Cheap Bootstrap produces wider confidence intervals at $B = 1$ than the other two methods, with an average interval width of 2.21. In this setting, the interval width also decreases sharply to 0.87 when increasing the number of bootstrap replications from $B = 1$ to $B = 2$. However, compared to the normally distributed case, the Cheap Bootstrap produces smaller interval widths at $B = 1$ for Gamma-distributed data. For larger numbers of bootstrap replications, all three methods converge to the same value, as can be seen in Figure 4a.

For the sample median, the Cheap Bootstrap achieves an empirical coverage of 92% at $B = 1$, improving slightly as B increases and reaching 94.7% at $B = 100$. The Normal and Percentile methods require at least $B = 50$ to approach similar coverage levels. Regarding the average interval widths, the Cheap Bootstrap produces intervals with a width of 2.59 at $B = 1$, which decreases sharply to 1.04 at $B = 2$ and approaches 0.55 at $B = 100$. The Normal and Percentile Bootstrap methods yield narrower intervals as B increases; however, all three methods converge to the same value (see Figure 4b).

For the sample quantile, the Cheap Bootstrap achieves the lowest empirical coverage of the three statistics at $B = 1$, but coverage improves as the number of bootstrap replications increases. Nevertheless, the Cheap method still outperforms the Normal and Percentile methods for smaller numbers of replications, although the resulting intervals are wider and decrease as the number of replications increases (see Figures 3c and 4c).

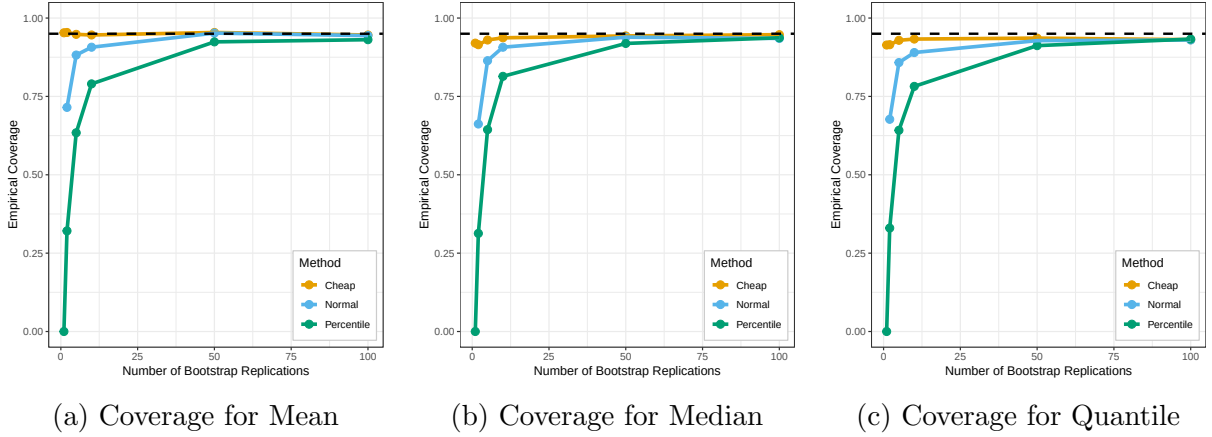


Figure 3: Empirical coverage of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the mean, median, and 0.75-quantile under Gamma-distributed data, based on 1000 simulated intervals for each value of B .

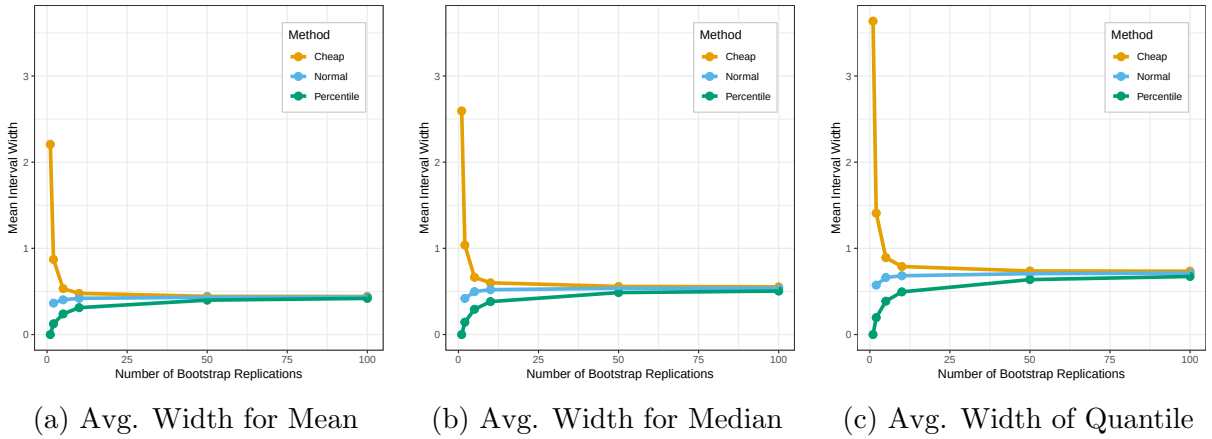


Figure 4: Average widths of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the mean, median, and 0.75-quantile under Gamma-distributed data, based on 1000 simulated intervals for each value of B .

4.3 Extreme Cases

In this section, we evaluate the robustness of the Cheap Bootstrap in extreme cases by considering the upper extreme quantile ($p = 0.99$) and a small sample size ($n = 20$), and we compare its performance with the Normal and Percentile Bootstrap methods.

4.3.1 Extreme Quantiles

For both normally and Gamma-distributed data, the Cheap Bootstrap achieves the highest empirical coverage across all values of B for the extreme upper 0.99-quantile when compared to the Normal and Percentile methods, yet the coverage does not reach the nominal 95%-level. The average interval widths for the Cheap Bootstrap are extremely wide for $B = 1$, but they quickly narrow as B increases. In the case of normally distributed

data, the empirical coverage of the Cheap Bootstrap increases from 63.8% to 80.6% as B grows, while the average interval width decreases from 11.27 at $B = 1$ to 2.51 at $B = 100$. The Normal Bootstrap attains coverage levels comparable to the Cheap Bootstrap for moderate values of B and above, reaching 70.8% at $B = 5$ and 77.9% at $B = 100$. In contrast, the Percentile Bootstrap remains noticeably below both the Cheap and Normal methods across all values of B , reaching only 62.5% at $B = 100$, while producing the narrowest intervals, whose widths increase from 0.56 to 1.78 (see Figure 5). A similar pattern is observed for Gamma distributed data. Empirical coverage of the Cheap Bootstrap increases from 62.8% to 81.0% as B grows, while the mean interval width decreases from 10.19 to 2.36. As in the normally distributed case, the Normal Bootstrap attains coverage levels comparable to the Cheap Bootstrap for small values of B . The Percentile Bootstrap remains considerably lower across all B , reaching only 64.6% at $B = 100$, while producing the narrowest intervals (see Figure 6).

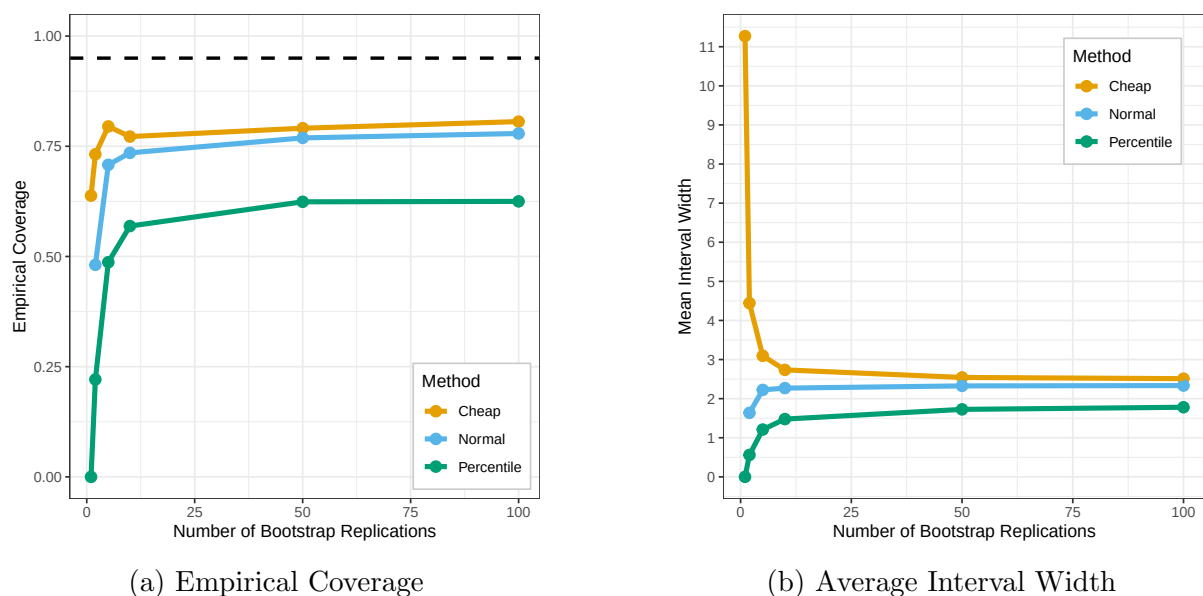


Figure 5: Empirical coverage and average interval widths of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the 0.99-quantile under normally distributed data ($n = 100$), based on 1000 simulated intervals for each value of B .

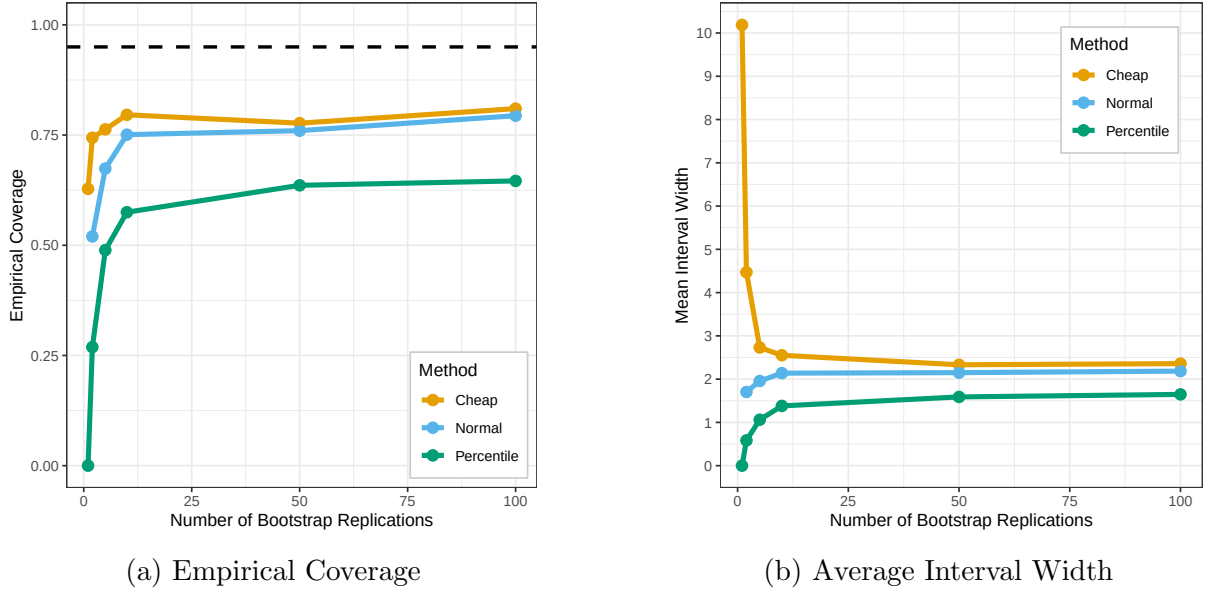


Figure 6: Empirical coverage and average interval widths of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the 0.99-quantile under Gamma-distributed data ($n = 100$), based on 1000 simulated intervals for each value of B .

4.3.2 Small Data Size

We now investigate how reducing the sample size to $n = 20$ affects the coverage and width of the bootstrap confidence intervals, allowing us to assess the robustness of the methods. For the sample mean, the Cheap Bootstrap performs well even with small sample sizes, achieving empirical coverage close to the nominal 95% level for both simulated datasets. However, the coverage slightly decreases as the number of bootstrap replications grows (from 94.6% to 92.7% for the normally distributed data, and from 94.5% to 92.5% for the Gamma-distributed data). In contrast, the empirical coverage of the Normal and Percentile Bootstrap methods increases with the number of bootstrap replications. The Normal Bootstrap reaches over 90% coverage already at $B = 10$, while the Percentile Bootstrap achieves this level only at $B = 50$ (see Figures 7a and 9a). Regarding the average interval widths for $B = 1$, the Cheap Bootstrap produces much wider intervals for smaller sample sizes than for larger sample sizes. For normally distributed data, the width is approximately 8.78 for $n = 20$ and 3.87 for $n = 100$. For Gamma-distributed data, the widths are 4.74 for $n = 20$ and 2.21 for $n = 100$. The average interval width of the Normal and Percentile Bootstraps is also wider for reduced sample sizes (see Figures 8a and 10a).

For the median, the Cheap Bootstrap empirical coverage at $B = 1$ is lower than that for the mean but is still higher than the coverage of the Normal and Percentile methods. The coverage for the Cheap Bootstrap improves with an increasing number of bootstrap replications, in contrast to the mean, for which coverage slightly decreases as B increases. The exact values for the three methods can be found in Tables 9 and 11. The interval widths for the median are also considerably wider for $B = 1$ compared to the mean.

The intervals for the Normal and Percentile methods are also wider than those for larger sample sizes, though they remain narrower than those of the Cheap Bootstrap.

Similar to the median, for the 0.75-quantile, the Cheap Bootstrap achieves the highest empirical coverage among the three methods for both normally and Gamma-distributed data. For $B = 1$, the coverage is slightly lower than for the median (85.9% for normal and 84.6% for gamma), and the interval widths are the widest among all statistics (about 11.43 for normal and 7.67 for gamma). The widths decrease sharply at $B = 2$, following the same pattern observed in previous results.

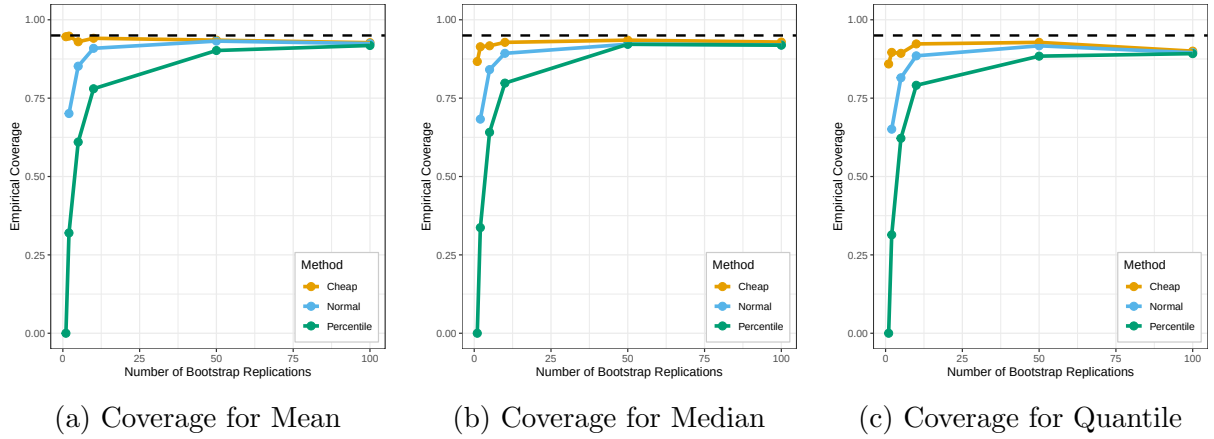


Figure 7: Empirical coverage of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the mean, median, and 0.75-quantile under normally distributed data with sample size $n = 20$, based on 1000 simulated intervals for each value of B .

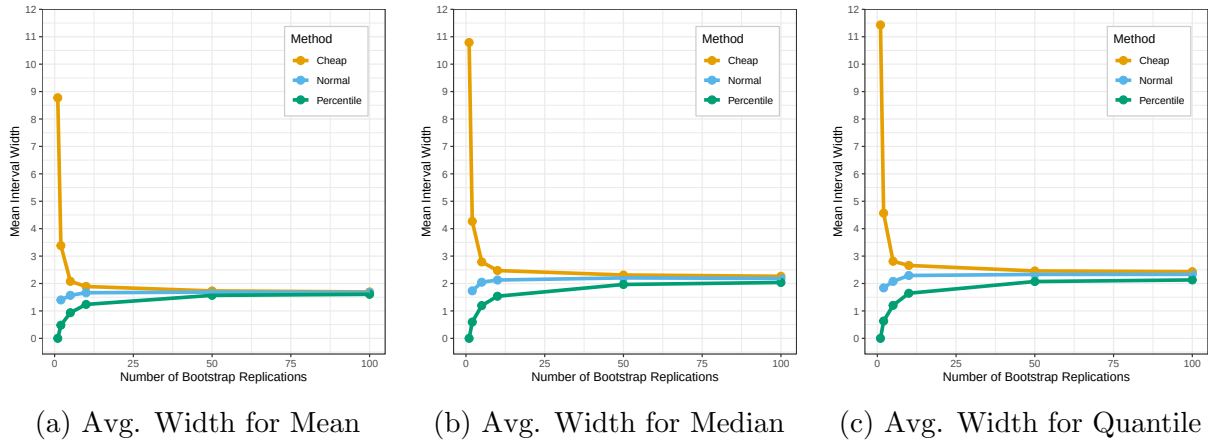


Figure 8: Average widths of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the mean, median, and 0.75-quantile under normally distributed data with sample size $n = 20$, based on 1000 simulated intervals for each value of B .

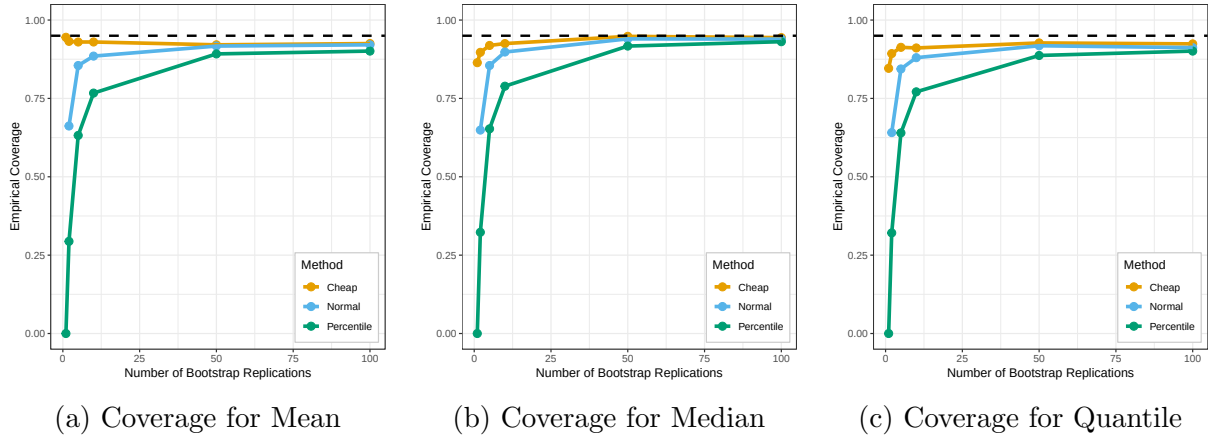


Figure 9: Empirical coverage of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the mean, median, and 0.75-quantile under Gamma-distributed data with sample size $n = 20$, based on 1000 simulated intervals for each value of B .

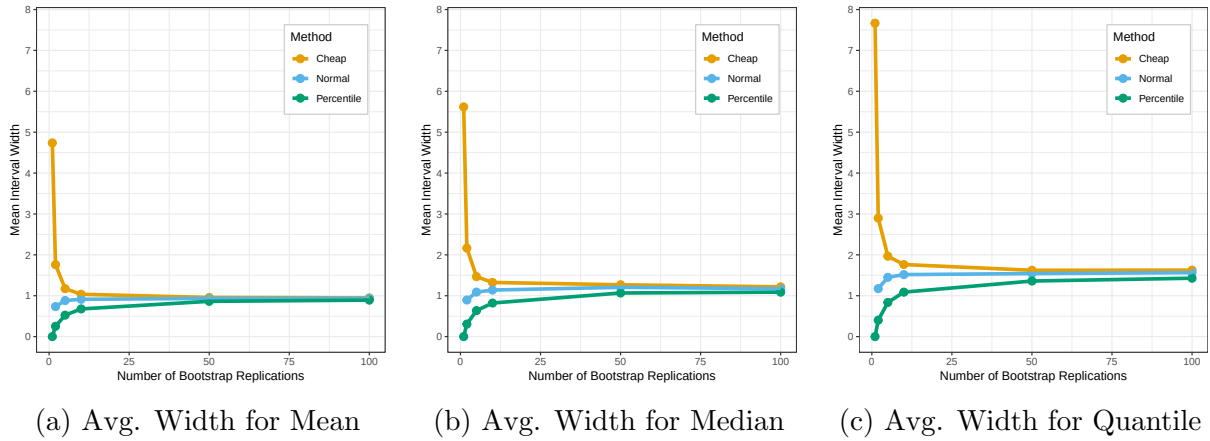


Figure 10: Average widths of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the mean, median, and 0.75-quantile under Gamma-distributed data with sample size $n = 20$, based on 1000 simulated intervals for each value of B .

4.4 Classical Linear Regression Model

In this section, we examine the performance of the Cheap Bootstrap in the context of a linear regression model and compare its empirical coverage with that of the two previously considered methods.

4.4.1 Theory of Linear Regression

We use regression models to study the relationship between a response variable y and one or more explanatory variables x , also referred to as covariates. Specifically, we aim to quantify the effect of the explanatory variables on the response variable y . A common and simple way to represent this relationship is the classical linear regression model. In the case of a single explanatory variable, the model is written as

$$y = \beta_0 + \beta_1 x + \varepsilon, \quad (33)$$

where β_0 is the intercept, β_1 the slope parameter, and ε the random error term. More generally, when multiple explanatory variables are considered, the linear regression model is defined as

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \varepsilon, \quad (34)$$

where $x_1, \dots, x_k$ denote the covariates of interest and $\beta_1, \dots, \beta_k$ their corresponding coefficients. The coefficients are unknown parameters that must be estimated from the data. To estimate them, we need observations for both the explanatory variables and the response variable. We generate n data points (x_i, y_i) , $i = 1, \dots, n$. For each observation i , we construct the following equation

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (35)$$

In the case of multiple linear regression, where each observation i has multiple explanatory variables $x_i = (x_{i1}, \dots, x_{ik})'$, the equation is then written as

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i. \quad (36)$$

This yields n equations, which can be summarized in vector notation as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (37)$$

where

$$\mathbf{y} = (y_1, \dots, y_n)', \quad \boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_k)', \quad \boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)',$$

and the design matrix is given by

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix}.$$

The unknown parameter vector β is estimated using the method of least squares, which determines the coefficients by minimizing the sum of squared deviations

$$LS(\beta) = \sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 \quad (38)$$

with respect to $\beta \in \mathbb{R}^{k+1}$. Minimizing $LS(\beta)$ yields the following least squares estimator $\hat{\beta}$:

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^{k+1}} LS(\beta) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}, \quad (39)$$

provided that $\mathbf{X}'\mathbf{X}$ is positive definite and invertible.

4.4.2 Simulation Setup

To assess the empirical coverage and average interval widths of confidence intervals for the regression parameters β , we simulate a multiple linear regression model with $k = 5$ explanatory variables and sample size $n = 100$. For each covariate X_j , $j = 1, \dots, k$, the observations $(x_{1j}, \dots, x_{nj})$ are independently drawn from a normal distribution with mean 10 and standard deviation 2. The regression coefficients β_j are fixed and known in advance, with all coefficients set equal to one, i.e., $\beta_j = 1$ for all $j = 1, \dots, k$, while the intercept parameter satisfies $\beta_0 = 0$. The error terms ε_i are independently drawn from a normal distribution with mean 0 and standard deviation 5. The response variable y_i is then computed according to the multiple linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \varepsilon_i, \quad i = 1, \dots, n. \quad (40)$$

As in our previous simulations, we construct 95% confidence intervals for the regression coefficients using the three bootstrap methods. In addition, we compute the classical confidence intervals obtained from the linear regression model using the `confint()` function in R, which serve as a baseline for comparison. It is important to note that these LM confidence intervals are not derived from the bootstrap samples. Instead, they are calculated directly from the original dataset.

4.4.3 Results

Since all covariates are generated independently from the same normal distribution and all regression coefficients share the same true value, the resulting empirical coverages and average interval widths are highly similar across all coefficients. We therefore focus on the results for the regression coefficient β_1 to avoid unnecessary redundancy and maintain conciseness. The corresponding results for the remaining coefficients are reported in Table A.5.4.

From Figure 11a, we can observe that the Cheap Bootstrap confidence intervals achieve the highest empirical coverage among the three bootstrap methods for all values of B . For $B = 1$, the Cheap Bootstrap achieves 94.7% coverage, which is slightly higher than the 94.6% achieved by the confidence intervals computed from the original linear model using `confint()`. As B increases, the coverage of the Cheap Bootstrap intervals stabilizes

at the nominal 95% level. In contrast, the Normal Bootstrap intervals start at 71.6% coverage for $B = 2$ and require 10 bootstrap replications to exceed 90% coverage. The Percentile Bootstrap intervals start even lower at 32.4% coverage for $B = 2$ and only reach comparable coverage at $B = 50$.

The Cheap Bootstrap produces the widest average intervals. At $B = 1$, the mean interval width is 5.14, but it decreases sharply to 1.97 for $B = 2$. The Cheap intervals only reach a width comparable to that of the classical linear model confidence intervals (LM) at around $B = 50$ replications. Compared to the other two methods, the Percentile Bootstrap produces the narrowest average intervals, starting at 0.28 for $B = 2$ and increasing to 0.97 for $B = 100$. The Normal Bootstrap intervals lie in between, with an average width of 0.81 at $B = 2$ and increasing to 1.01 for $B = 100$.

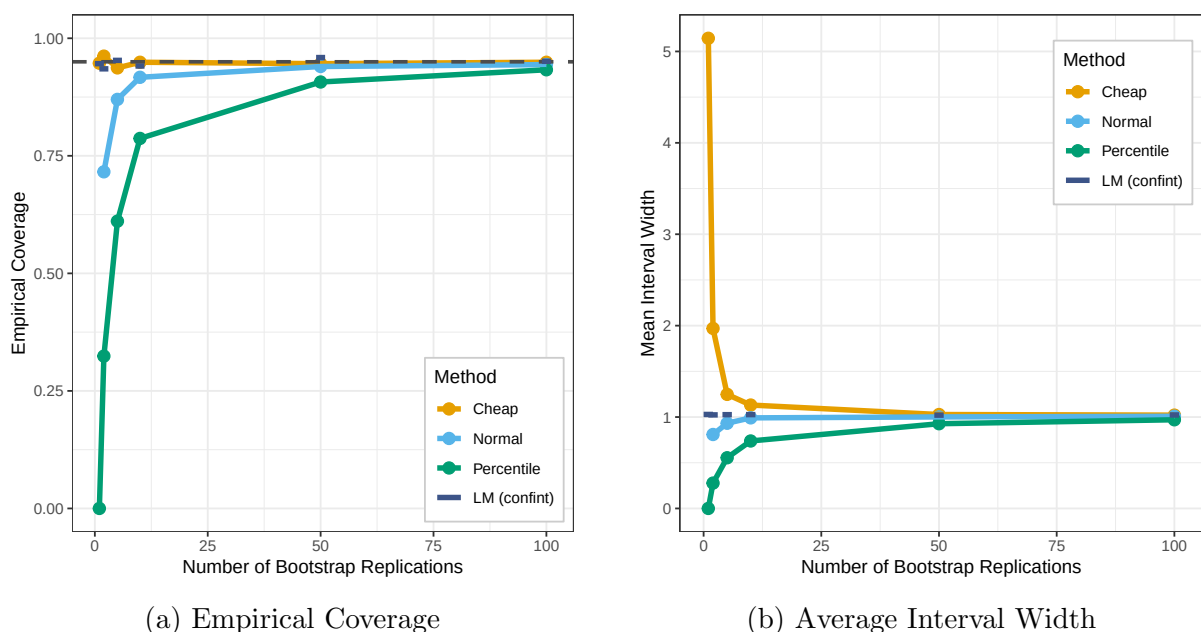


Figure 11: Empirical coverage and average interval widths of 95% confidence intervals constructed using the Normal, Percentile, and Cheap Bootstrap methods for the regression coefficient β_1 . Empirical coverage and interval widths from the classical linear regression model (LM) using `confint()` are included as a baseline for comparison. Results are based on 1000 simulated intervals for each value of B .

5 Conclusion

This thesis investigated the performance of the Cheap Bootstrap confidence intervals through an extensive simulation study. Datasets were generated from both Normal and Gamma distributions, and confidence intervals for a variety of parameters were constructed using the Cheap Bootstrap as well as the Normal and Percentile Bootstrap methods. Their performance was evaluated in terms of empirical coverage and average interval width, including challenging scenarios such as small sample sizes, extreme quantiles, and multiple linear regression models. Overall, the simulation study demonstrates that the Cheap Bootstrap confidence intervals provide valid empirical coverage across different parameters and datasets. Coverage was generally close to the nominal 95%-level even with minimal bootstrap replications, and the Cheap Bootstrap typically achieved the highest empirical coverage among the three methods. In terms of interval width, Cheap Bootstrap intervals were initially very wide for a single replication, but decreased sharply with just one additional replication and gradually approached the widths of the conventional methods as the number of replications increased. The robustness of the Cheap Bootstrap intervals was further assessed under challenging conditions. For extreme quantiles, Cheap Bootstrap generally achieved the highest empirical coverage, although none of the methods reached the nominal 95%-level. The Normal Bootstrap performed similarly, while the Percentile Bootstrap was the least accurate. For small samples ($n = 20$), Cheap Bootstrap intervals remained relatively stable and provided near-nominal coverage for the mean, but slightly underperformed for the median and upper quantiles until more replications were used. In the multiple linear regression setting, the results were consistent with the previous simulation studies: Cheap Bootstrap attained the highest empirical coverage, followed by Normal and Percentile Bootstrap. The average interval widths for Cheap Bootstrap followed the same pattern, being wide for a single replication ($B = 1$) and decreasing sharply with just one additional replication.

From these results, it could be argued that the Cheap Bootstrap attains high empirical coverage for $B = 1$ primarily due to the use of very wide confidence intervals. In fact, it is more likely that the true parameter value lies within a confidence interval when the interval is wider, which naturally leads to a higher value of empirical coverage. However, the simulation results show that this explanation is only partially true. As the number of bootstrap replications increases, the width of the Cheap Bootstrap confidence intervals decreases substantially, while empirical coverage does not decrease but instead remains stable or even improves. This illustrates a balance between interval width and empirical coverage, where wider intervals naturally lead to higher coverage; however, with more bootstrap replications, the Cheap Bootstrap method is able to reduce the widths without compromising coverage, effectively achieving a favorable trade-off between precision and reliability. Consequently, using only a single bootstrap replication ($B = 1$) already achieves high coverage and makes it suitable for quick and reliable estimates. Increasing the number of replications allows for narrower, more precise intervals without sacrificing coverage, although this comes at the expense of longer computation time (Lam, 2022). Overall, the results suggest that the Cheap Bootstrap provides a robust and computationally efficient alternative to commonly used bootstrap confidence interval methods, such as the Normal or Percentile Bootstrap. It is particularly well suited for situations in which

computational resources are limited or only a small number of bootstrap replications are feasible.

A Appendix

A.1 Use of AI Tools

- The core structure and basic code implementation were developed independently by me and subsequently extended and improved with the support of ChatGPT, with all suggestions critically reviewed before adoption.
- The organization and structure of the GitHub project were refined with the support of ChatGPT.
- The code for the visualizations in the figures section was largely created by ChatGPT according to my specifications and adopted after thorough review.
- I hereby confirm that the content and ideas of this thesis were independently developed by me. I wrote initial drafts and formulated all core concepts myself. ChatGPT, Claude, and Grammarly were then used to refine the language, grammar, spelling, and overall text flow of these drafts. All suggestions were critically reviewed and adapted by me. Final responsibility for all content lies with the author.

A.2 Convergence of Random Variables

Let $X_1, X_2, \dots$ be random variables.

Convergence in distribution: $X_n \xrightarrow{d} X \Leftrightarrow P(X_n \leq x) \rightarrow P(X \leq x) \text{ as } n \rightarrow \infty$

Convergence in probability: $X_n \xrightarrow{P} X \Leftrightarrow P(|X_n - X| > \varepsilon) \rightarrow 0 \forall \varepsilon > 0, \text{ as } n \rightarrow \infty$

Almost sure convergence: $X_n \xrightarrow{a.s.} X \Leftrightarrow P\left(\lim_{n \rightarrow \infty} |X_n - X| = 0\right) = 1$

A.3 Continuous Mapping Theorem

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Then:

$$\begin{aligned} X_n \xrightarrow{d} X &\implies g(X_n) \xrightarrow{d} g(X) \\ X_n \xrightarrow{P} X &\implies g(X_n) \xrightarrow{P} g(X) \\ X_n \xrightarrow{a.s.} X &\implies g(X_n) \xrightarrow{a.s.} g(X) \end{aligned}$$

A.4 Law of Large Numbers

Let $X_1, \dots, X_n \sim F$ be i.i.d. random variables.

Weak Law of Large Numbers: If $\mathbb{E}|g(X_1)| < \infty$, then

$$\frac{1}{n} \sum_{i=1}^n g(X_i) \xrightarrow{P} \mathbb{E}[g(X_1)] \text{ as } n \rightarrow \infty.$$

Strong Law of Large Numbers: If $\mathbb{E}|g(X_1)| < \infty$, then

$$\frac{1}{n} \sum_{i=1}^n g(X_i) \xrightarrow{a.s.} \mathbb{E}[g(X_1)] \quad \text{as } n \rightarrow \infty.$$

A.5 Tabular Results

A.5.1 Bootstrap Performance for $n = 100$

Normally Distributed Data ($n = 100$)

Table 1: Empirical coverage for normally distributed data, $n = 100$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB			NB			PB		
	Mean	Median	Quantile	Mean	Median	Quantile	Mean	Median	Quantile
1	0.946	0.909	0.901	—	—	—	0.000	0.000	0.000
2	0.947	0.933	0.903	0.679	0.687	0.646	0.315	0.317	0.316
5	0.959	0.924	0.925	0.881	0.847	0.859	0.642	0.661	0.631
10	0.931	0.934	0.925	0.900	0.903	0.897	0.774	0.782	0.773
50	0.947	0.942	0.950	0.941	0.940	0.943	0.910	0.916	0.925
100	0.939	0.942	0.932	0.935	0.936	0.925	0.922	0.931	0.929

Table 2: Mean interval widths for normally distributed data, $n = 100$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB			NB			PB		
	Mean	Median	Quantile	Mean	Median	Quantile	Mean	Median	Quantile
1	3.867	5.047	5.666	—	—	—	0.000	0.000	0.000
2	1.532	1.970	2.097	0.644	0.797	0.847	0.221	0.273	0.290
5	0.956	1.228	1.348	0.725	0.918	1.002	0.431	0.540	0.588
10	0.858	1.117	1.221	0.755	0.968	1.056	0.561	0.707	0.770
50	0.794	1.024	1.131	0.775	0.985	1.082	0.719	0.893	0.977
100	0.785	1.009	1.124	0.776	0.981	1.092	0.745	0.925	1.019

Gamma Distributed Data ($n = 100$)Table 3: Empirical coverage for gamma distributed data, $n = 100$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB			NB			PB		
	Mean	Median	Quantile	Mean	Median	Quantile	Mean	Median	Quantile
1	0.953	0.920	0.914	–	–	–	0.000	0.000	0.000
2	0.954	0.915	0.915	0.715	0.662	0.677	0.321	0.313	0.330
5	0.948	0.930	0.929	0.882	0.864	0.858	0.634	0.644	0.642
10	0.946	0.937	0.933	0.907	0.907	0.890	0.790	0.814	0.782
50	0.954	0.943	0.936	0.952	0.939	0.928	0.924	0.919	0.912
100	0.946	0.947	0.931	0.944	0.935	0.930	0.931	0.937	0.933

Table 4: Mean interval widths for gamma distributed data, $n = 100$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB			NB			PB		
	Mean	Median	Quantile	Mean	Median	Quantile	Mean	Median	Quantile
1	2.207	2.594	3.637	–	–	–	0.000	0.000	0.000
2	0.872	1.037	1.408	0.364	0.418	0.573	0.125	0.143	0.196
5	0.532	0.664	0.893	0.403	0.498	0.662	0.239	0.293	0.387
10	0.478	0.600	0.789	0.420	0.520	0.682	0.312	0.382	0.495
50	0.443	0.558	0.738	0.432	0.536	0.706	0.399	0.487	0.637
100	0.442	0.553	0.735	0.436	0.538	0.713	0.419	0.504	0.674

A.5.2 Bootstrap Performance for Extreme Quantiles

Normally Distributed Data, 99th Percentile ($n = 100$)

Table 5: Empirical coverage for the 99th percentile, $n = 100$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB Quantile	NB Quantile	PB Quantile
1	0.638	—	0.000
2	0.732	0.481	0.221
5	0.795	0.708	0.487
10	0.772	0.735	0.569
50	0.791	0.769	0.624
100	0.806	0.779	0.625

Table 6: Average interval widths for the 99th percentile, $n = 100$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap. Values rounded to 3 decimals.

B	CB Quantile	NB Quantile	PB Quantile
1	11.272	—	0.000
2	4.442	1.637	0.561
5	3.096	2.225	1.209
10	2.736	2.270	1.477
50	2.542	2.326	1.726
100	2.511	2.335	1.780

Gamma Distributed Data, 99th Percentile ($n = 100$)

Table 7: Empirical coverage for the 99th percentile, $n = 100$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB Quantile	NB Quantile	PB Quantile
1	0.628	–	0.000
2	0.744	0.520	0.269
5	0.763	0.674	0.489
10	0.796	0.751	0.575
50	0.777	0.760	0.636
100	0.810	0.794	0.646

Table 8: Average interval widths for the 99th percentile, $n = 100$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB Quantile	NB Quantile	PB Quantile
1	10.188	–	0.000
2	4.471	1.704	0.584
5	2.731	1.957	1.062
10	2.551	2.138	1.381
50	2.332	2.149	1.589
100	2.358	2.185	1.647

A.5.3 Bootstrap Performance for Small Sample Sizes

Normally Distributed Data ($n = 20$)

Table 9: Empirical coverage for normally distributed data, $n = 20$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB Width			NB Width			PB Width		
	Mean	Median	Quantile	Mean	Median	Quantile	Mean	Median	Quantile
1	0.946	0.867	0.859	—	—	—	0.000	0.000	0.000
2	0.948	0.914	0.896	0.701	0.683	0.651	0.320	0.337	0.314
5	0.930	0.917	0.893	0.852	0.841	0.815	0.610	0.641	0.622
10	0.941	0.928	0.923	0.909	0.893	0.885	0.780	0.798	0.791
50	0.935	0.935	0.928	0.932	0.922	0.917	0.902	0.922	0.884
100	0.927	0.929	0.900	0.924	0.920	0.895	0.918	0.919	0.892

Table 10: Mean interval widths for normally distributed data, $n = 20$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB			NB			PB		
	Mean	Median	Quantile	Mean	Median	Quantile	Mean	Median	Quantile
1	8.775	10.791	11.432	—	—	—	0.000	0.000	0.000
2	3.382	4.267	4.565	1.400	1.734	1.839	0.480	0.594	0.630
5	2.080	2.787	2.811	1.569	2.047	2.077	0.935	1.195	1.203
10	1.891	2.473	2.658	1.664	2.127	2.292	1.237	1.536	1.645
50	1.730	2.311	2.458	1.686	2.204	2.332	1.568	1.965	2.073
100	1.693	2.264	2.431	1.673	2.185	2.333	1.606	2.038	2.130

Gamma Distributed Data ($n = 20$)Table 11: Empirical coverage for gamma distributed data, $n = 20$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB			NB			PB		
	Mean	Median	Quantile	Mean	Median	Quantile	Mean	Median	Quantile
1	0.945	0.864	0.846	–	–	–	0.000	0.000	0.000
2	0.932	0.897	0.893	0.662	0.649	0.641	0.294	0.323	0.321
5	0.930	0.919	0.913	0.855	0.855	0.844	0.632	0.653	0.640
10	0.930	0.925	0.911	0.885	0.898	0.880	0.767	0.789	0.771
50	0.921	0.948	0.927	0.917	0.940	0.918	0.892	0.917	0.887
100	0.925	0.944	0.924	0.921	0.939	0.912	0.901	0.932	0.901

Table 12: Mean interval widths for gamma distributed data, $n = 20$. CB: Cheap Bootstrap, NB: Normal Bootstrap, PB: Percentile Bootstrap.

B	CB			NB			PB		
	Mean	Median	Quantile	Mean	Median	Quantile	Mean	Median	Quantile
1	4.738	5.619	7.667	–	–	–	0.000	0.000	0.000
2	1.760	2.165	2.899	0.735	0.896	1.173	0.252	0.307	0.402
5	1.173	1.467	1.967	0.882	1.089	1.451	0.525	0.636	0.836
10	1.036	1.324	1.763	0.913	1.139	1.516	0.677	0.822	1.088
50	0.957	1.268	1.622	0.934	1.206	1.542	0.863	1.067	1.359
100	0.947	1.218	1.626	0.936	1.172	1.565	0.892	1.088	1.429

A.5.4 Bootstrap Performance for Multiple Linear Regression

Table 13: Empirical coverage and mean interval width for all regression coefficients using the Cheap Bootstrap method

B	β_1	β_2	β_3	β_4	β_5
Empirical Coverage					
1	0.947	0.946	0.950	0.954	0.958
2	0.962	0.949	0.935	0.955	0.953
5	0.937	0.959	0.952	0.942	0.948
10	0.949	0.949	0.950	0.931	0.947
50	0.946	0.944	0.942	0.947	0.948
100	0.949	0.945	0.951	0.948	0.946
Mean Interval Width					
1	5.144	5.154	5.328	5.100	5.322
2	1.969	1.974	1.938	1.989	1.961
5	1.248	1.255	1.260	1.266	1.263
10	1.131	1.128	1.115	1.117	1.120
50	1.028	1.016	1.027	1.031	1.022
100	1.021	1.013	1.017	1.011	1.013

Table 14: Empirical coverage and mean interval width for all regression coefficients using the Normal Bootstrap method

B	β_1	β_2	β_3	β_4	β_5
Empirical Coverage					
1	—	—	—	—	—
2	0.716	0.690	0.694	0.696	0.699
5	0.870	0.883	0.880	0.874	0.862
10	0.917	0.913	0.912	0.899	0.911
50	0.940	0.935	0.940	0.941	0.940
100	0.944	0.942	0.946	0.946	0.942
Mean Interval Width					
1	—	—	—	—	—
2	0.809	0.815	0.822	0.809	0.806
5	0.932	0.952	0.954	0.959	0.952
10	0.991	0.988	0.978	0.980	0.980
50	1.001	0.991	1.001	1.006	0.996
100	1.008	1.000	1.004	0.999	1.000

Table 15: Empirical coverage and mean interval width for all regression coefficients using the Percentile Bootstrap method

B	β_1	β_2	β_3	β_4	β_5
Empirical Coverage					
1	0.000	0.000	0.000	0.000	0.000
2	0.324	0.321	0.288	0.310	0.336
5	0.611	0.656	0.657	0.646	0.619
10	0.787	0.757	0.778	0.778	0.797
50	0.907	0.906	0.917	0.898	0.902
100	0.933	0.925	0.938	0.929	0.932
Mean Interval Width					
1	0.000	0.000	0.000	0.000	0.000
2	0.277	0.279	0.282	0.277	0.276
5	0.555	0.568	0.568	0.572	0.566
10	0.738	0.733	0.728	0.729	0.730
50	0.927	0.917	0.925	0.932	0.921
100	0.970	0.961	0.966	0.959	0.964

Table 16: Empirical coverage and mean interval width for all regression coefficients using linear model confidence intervals. Note that these intervals were computed with `confint()` and do not depend on the number of bootstrap replications B .

B	β_1	β_2	β_3	β_4	β_5
Empirical Coverage					
1	0.946	0.959	0.946	0.943	0.954
2	0.935	0.958	0.949	0.957	0.951
5	0.953	0.956	0.950	0.948	0.946
10	0.941	0.950	0.948	0.941	0.934
50	0.959	0.957	0.952	0.949	0.947
100	0.951	0.956	0.952	0.955	0.951
Mean Interval Width					
1	1.028	1.021	1.024	1.027	1.022
2	1.024	1.026	1.025	1.026	1.021
5	1.026	1.027	1.026	1.025	1.024
10	1.027	1.026	1.027	1.028	1.023
50	1.021	1.018	1.021	1.020	1.020
100	1.023	1.022	1.020	1.022	1.019

B Electronic appendix

Data, code and figures are provided in electronic form and are available at: <https://github.com/CongHungEissrig/Cheap-Bootstrap>

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